

So AI harms the environment? How?

Posted by u/ItWasMineFirst • 0 points • 15 comments

So a good friend of mine says that AI harms the planet. I went on chat GPT while at his house and he was like "you know one conversation with GPT is like tossing a plastic water bottle in the sea" and I was like ???

How does GPT or any other AI harm the planet more than anything else on the web or in technology? (This guy is a big computer buff). Am I missing some kind of recent information or something

Comments

u/rhomboidus • 2 points • 2025-01-02 14:04 UTC

Asking ChatGPT what the weather is burns a tremendous amount of electricity when you could just look out a window instead.

u/HopeSubstantial • 2 points • 2025-01-02 14:03 UTC

AI uses insane amount of electricity. Google is planning to build nuclear powerplant for AI alone.

Currently sadly alot of countries run their AI with dirty power.

u/Clojiroo • 2 points • 2025-01-02 13:54 UTC

It's important to note that:

- this is about LLMs, not all machine learning or AI
- this is very model-dependent. The hardware required to power a model like 4o is very different from say, a model that can run on your phone or

The hardware required to power advanced models is substantial. They are enormous. It uses a lot of electricity. A **lot**. Think of it like (way over simplified) millions of extra gaming PCs running so you can ask copilot how to rewrite your memo.

Cloud providers are trying to quickly ramp up energy production. They need power plants.

u/ChemicalTerrapin • 3 points • 2025-01-02 13:51 UTC

As others have said, it's about the amount of compute required.

A couple of extra bits of fun information....

1. A lot of machine learning development is done in python. There's no great reason for that other than data scientists came from that background and that's how it has stayed.

But it's roughly 70 times less energy efficient than C for the same task.

2. This is one of the reasons why Amazon, MS and Google are investing so heavily in small scale nuclear power. We are very quickly outgrowing our energy consumption limits.

u/ItWasMineFirst • 3 points • 2025-01-02 13:52 UTC

Thank you for the response!

u/SiewcaWiatru • 5 points • 2025-01-02 13:48 UTC

AI computing uses way more computing power than any other stuff. Just that.

It's not AI thing as in thinking or results but the power it requires to work. Computing answers to questions and huge storage facilities require a lot more.

Loads of different PUs, storage for persistent data, RAM for in-memory stuff etc.

u/frizzykid • 1 points • 2025-01-02 13:44 UTC

>"you know one conversation with GPT is like tossing a plastic water bottle in the sea" and I was like ???

This is a ridiculous comparison and makes very little sense. but developing Ai is significantly more power intensive than most other computing operations we do. That isn't necessarily a bad thing it just means we need to diversify our energy usage which many Ai developing companies are doing with nuclear.

u/Pastadseven • 2 points • 2025-01-02 13:52 UTC

It's especially annoying because those power concerns are a real fucking problem and it's butied under clickbait bullshit about bottles. It'd be *nice* if nuclear made up the shortfall but in reality given the locations of data centers, there's no nuclear infrastructure, it's just going to be more fossil fuel burning. West Virginia? Good fucking luck. There are zero nuclear power plants there. The power's coming from coal.

u/ItWasMineFirst • 2 points • 2025-01-02 13:51 UTC

That's funny you say "something you'd read in a sensationalised news article against AI" as hes actually a writer for techradar and has made some anti-AI articles before lol.

I guess he's just very biased against it. I'm in no way a tech expert but I've found AI to be extremely helpful to my every day life and I don't really understand why people have such an issue with it (other than of course when AI generated images etc are passed as authentic art etc)

u/Pastadseven • 1 points • 2025-01-02 14:04 UTC

Well, there's the other side where a lot of AI-generated anything is just fucking garbage. Especially the writing.

u/No-Mushroom5934 • 1 points • 2025-01-02 13:44 UTC

yeahh they demand massive energy. servers, data centers, even this conversation, all need electricity and usually from sources that degrade the planet over time. it is kinda hidden

and that power, in maximum doses comes from burning things. just like tossing a plastic bottle into the sea as u said , it is not the bottle itself , it is the damage it silently causes. we not talking about just one bottle , we are talking about millions of invisible plastic bottles , they are silent contributors to an unseen disaster. it is not just data , it is what goes into producing that data. and that goes ignored...

u/MontCoDubV • 3 points • 2025-01-02 13:43 UTC

AI takes a LOT more computing power than most things most people use their computers for. More computing power requires a greater consumption of electricity. That electricity is almost certainly generated by burning fossil fuels.

AI increases the demand for electricity, which increases how much electricity we generate, which puts more carbon into the climate.

u/Pastadseven • 3 points • 2025-01-02 13:43 UTC

AI and especially crypto use a shitload of power. Like a footprint the size of a small country's worth of power.

u/jeffgroves • 14 points • 2025-01-02 13:39 UTC

AI apparently requires more electricity than things like search engines to answer a given question. It's similar to cryptocurrency mining in that regard: high CPU/GPU usage = uses more electricity

u/cearrach • 1 points • 2025-01-02 14:02 UTC

Just to expand on this a bit since some people are still missing the point...

Electricity generation is inherently polluting. "Throwing a bottle into the ocean" is being compared to something like "burning a bunch of coal" to generate electricity in that both are examples of pollution.

u/Lumpy-Notice8945 • -1 points • 2025-01-02 14:00 UTC

No, its not similar to crypto, crypto mining is like 100× times more energy usage!

Crypto mining is guessing random numbers over and over untill someone gets it right(proof of work)

AI runs on expensive but somehow regular GPUs, mining is done by millions of GPUs together by people all over the world, thats why its consuming as much power as a country!